EXCLUSIVE EDITION | 2026

By Sreemanti Ghosh — AI/ML Engineer

v1.0 — INDIA & WORLDWIDE EDITION

The AI Engineer Blueprint

A 4-Month Roadmap to Land a $100K+ AI/ML Engineer Role in 2026

ROADMAP | RESOURCES | RESUME | PROJECTS

From IIT Delhi → Google → TurboML → AI/ML Engineer at a Top US Startup

ABOUT THE AUTHOR

Hey, I'm Sreemanti.

I'm an **AI/ML Engineer working remote at a top US startup**. I graduated from **IIT Delhi (B.Tech, Computer Science) with Institute Rank 3**, and along the way I've interned at **Google** as a software engineer, worked at **Alphagrep Securities**, and broke into ML at **TurboML** — my first ML internship, which I landed because a recruiter saw a project I posted on LinkedIn.

I've been deep in machine learning for over two years now. I've interviewed for ML Engineer, AI Engineer, and AI Research Engineer roles. I've mentored hundreds of people on Topmate. And I've watched smart, hardworking people get rejected from AI/ML jobs not because they lacked talent, but because **nobody told them what to actually focus on**.

WHY I WROTE THIS

AI Engineer is the highest-paying job in the market right now, and the gap between what companies need and what most candidates have is painstakingly huge. I've researched Coursera, Stanford, Udemy, Krish Naik, 3Blue1Brown, CampusX, Educative.io, NPTEL — every major course online. This ebook is the **distilled, no-fluff path** through all of it. Follow it for 4 months and you will land the role.

What's inside:

- A **week-by-week, 4-month roadmap** with curated resources for every topic.
- A vault of **100% free** courses, books, projects, and interview question banks.
- A **resume teardown** — section by section — that helped me land my role.
- **14 recruiter-magnet projects** rated for difficulty, learning value, and resume value.
- My personal advice on **LinkedIn, posting, cold emails, and visibility**.

"If you're happy, I'm happy. I'm here to see you succeed."

— Sreemanti

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READ THIS FIRST

The 4-Month Promise

Let me get straight to the point. AI Engineer is the **highest-paying entry-level job in tech in 2026** — and it's still wide open. Every week I see companies on LinkedIn, Wellfound, and YC's job board posting AI roles they can't fill. The gap between the skills companies need and the skills candidates have is enormous.

This is good news for you. It means that with the right roadmap, the right resources, and the right projects on your resume, **you can break in** — even as a fresher, even from a non-tier-1 college, even on a budget.

THE PROMISE

Follow this roadmap for 4 months — really follow it, not skim it — and at the end you'll be able to deploy a FastAPI AI service, fine-tune an open model for a real use case, containerize it with Docker, serve it with Ollama or vLLM, monitor it, and publish the whole thing to GitHub and LinkedIn. **That is the project that gets you hired immediately.**

How to use this ebook

The roadmap is sequential — please don't skip ahead. Month 2 builds on Month 1, Month 3 builds on Month 2, and so on. Inside each month, the topics are also ordered. You'll find:

- **Goal** — what you'll be able to do by the end of the month.
- **Topics** — exactly what to learn and the resource that teaches it best.
- **Projects** — what to build to make it stick (and put on your resume).
- **Action items** — small concrete steps for the week.

"Visibility matters more than you think. Build projects — but don't stop there. Post them. Share them. Let the world know what you're building."

MONTH

1

Foundations

Python, Math, Data & Classical ML

The goal of Month 1

Build rock-solid Python fundamentals, learn the math *intuition* behind ML, get comfortable with the core AI engineering tooling, understand the ML lifecycle, and ship your first machine learning project. It's intensive — but if you want the highest-paying job in tech and security in your career, this is non-negotiable.

Week 1 · Python that actually works

Start with **Programming with Mosh — Python Full Course (6 hrs)**. That single video is enough for fundamentals. Don't binge it; code along. By the end of the week you should be confident with functions, classes, list/dict comprehensions, file I/O, and virtual environments.

Week 2 · The ML Python stack

Learn just enough of NumPy, Pandas, Matplotlib, and scikit-learn to be dangerous. Find a 10-min-per-video playlist for each — that depth is exactly what AI Engineer interviews test. Don't go deeper than that until you've shipped a project.

> **RULE OF THUMB**
>
> If a YouTube video is longer than 30 minutes for a single library, scroll past it. Short, focused tutorials win. You'll fill in the gaps when you build.

Week 3 · Math intuition (not proofs)

You don't need to re-derive eigenvalues. You need **intuition**. Pick:

Stanford Probability & Statistics — first 6 videos — youtube.com/playlist?list=PLoROMvodv4rOpr_A7B9SriE_iZmkanvUg

3Blue1Brown — Linear Algebra & Calculus playlists — youtube.com/c/3blue1brown
Best visual intuition on the internet. 1–2 days, max.

Week 3 · Tooling: Git, GitHub, terminal

Pick any solid 1-hour Git/GitHub crash course. Learn branches, pull requests, merging, and pushing your work. Every project from here on lives on GitHub.

Week 4 · Classical ML with Andrew Ng

Coursera Machine Learning Specialization (Andrew Ng) — even the 2011 version is still relevant in 2026. Linear/logistic regression, decision trees, neural networks, and hands-on labs. This single course got me into Google when ML was in its infancy.

Free YouTube alternative — Stanford CS229 full playlist — youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU

Same material, no labs, completely free.

Project — your first ML build

> **BUILD: YouTube Video Summarizer**
>
> Pull a transcript with the YouTube Transcript API, chunk it, run it through a summarization model, and print a bulleted summary. It's beginner-friendly, it teaches you the real shape of an AI engineer's day, and it's a clean first repo on GitHub.

End-of-month checklist

- Comfortable with Python, NumPy, Pandas, Matplotlib, scikit-learn.
- Stat/probability/linear-algebra/calculus *intuition* in place.
- Git/GitHub muscle memory.
- Andrew Ng ML Specialization completed.
- First project shipped to GitHub *and* posted on LinkedIn.

MONTH

2

Going Deeper

Deep Learning, PyTorch, SQL & Internships

The goal of Month 2

Master neural networks, get comfortable with PyTorch, ship 1–2 deep learning projects to GitHub, and — this is critical — **start applying for internships now**. Yes, now. Even if you feel underprepared. Interviews will tell you exactly where you're weak, faster than any course can.

> **DO THIS WEEK 1: APPLY EVERYWHERE**
>
> AI Engineer, Data Analyst, even pure Python internships count. The fastest way to level up is to interview while you're still studying. Lose the pride and apply.

Where to apply

• **LinkedIn** — obvious, but use it daily and message recruiters.
• **Wellfound (formerly AngelList)** — wellfound.com — tons of US startups, remote-friendly, and arguably better signal than LinkedIn for early-career AI roles.
• **Cold email IIT/IIIT/IISc professors** — ask to join their ML/AI research lab as an intern. A 2–3 month research stint is a *massive* resume boost.
• **Kaggle competitions & hackathons** — they double as projects and as interview talking points.

Week 1–2 · Master SQL

Every ML role I've held has had a heavy SQL component. Don't skip this.

freeCodeCamp — full SQL course on YouTube — youtube.com/@freecodecamp

Gate Smashers SQL playlist — DB internals & query optimization — youtube.com/@GateSmashers
One of the best free SQL/DBMS resources in the world.

Week 2–3 · PyTorch fundamentals

Tensors, autograd, simple training loops, dataloaders. Pick any focused PyTorch YouTube tutorial — don't go deeper than autograd until you're building.

Week 3–4 · Deep Learning Specialization (Andrew Ng)

Get a real grip on backpropagation, optimizers, loss functions, CNNs, RNNs, and LSTMs. If you want it free, watch the YouTube version of Stanford CS230.

Stanford CS230 — Deep Learning (free, full playlist) — youtube.com/playlist?list=PLoROMvodv4rNRRGdS0rBbXOUGA0wjdh1X

Projects to ship this month

BUILD: Image Classification System (Cats vs. Dogs)

Train a CNN from scratch in PyTorch, then add transfer learning with ResNet. Compare results. Push to GitHub. Write a one-paragraph LinkedIn post about what you learned. This is the canonical first deep-learning project that recruiters still respect — if you put a non-trivial twist on it.

BONUS: Housing Price Predictor

Classic regression project — but treat it as a chance to practice the full ML lifecycle: EDA, feature engineering, model selection, hyperparameter tuning, evaluation. Document your decisions in the README. Recruiters read READMEs.

End-of-month checklist

- Comfortable with PyTorch — tensors, autograd, training loops.
- CNNs, RNNs/LSTMs, optimizers, loss functions explained in your own words.
- SQL queries written without Googling.
- 2 deep learning projects on GitHub, posted on LinkedIn.
- **Applied to at least 25 internships.** Interviewing already.

MONTH

3

The LLM Era

Transformers, NLP, RAG & Vector DBs

The goal of Month 3

This is where you become an actual AI Engineer. Modern LLM workflows, tool calling, embeddings, transformers, prompt design, retrieval-augmented generation, fine-tuning, and the vector database stack. By the end of the month you'll be shipping real LLM-powered apps.

Week 1 · NLP foundations

NPTEL NLP — IIT Madras (Prof. Pawan Goyal) — nptel.ac.in/courses/106105158
Watch up to Module 7. Free, with assignments and a certificate option.

Week 1–2 · Tokenization, embeddings, transformers

You need an honest mental model of **how a transformer actually works** — attention, KV cache, positional encoding. Two resources:

3Blue1Brown — Visual guide to LLMs (playlist) — youtube.com/c/3blue1brown
The clearest visual intuition on the planet.

AwesomeNeuron — A Visual Guide to LLMs — awesomeneuron.substack.com/p/a-visual-guide-to-llms-part-1
Interactive visual deep-dive. Bookmark every post.

Week 2–3 · RAG, fine-tuning, tool calling

These are the topics that show up in *every* AI Engineer interview right now.

- **RAG** — chunking, embeddings, vector DB retrieval, reranking, evaluation.
- **Fine-tuning** — LoRA, QLoRA, when to fine-tune vs. when to RAG.
- **Tool calling** — function calling APIs, structured output, agent loops.
- **Vector DBs** — ChromaDB, FAISS, Pinecone basics.
- **Frameworks** — LangChain, LlamaIndex, CrewAI.

> **RECRUITER HOT BUTTON**
>
> Multiple students of mine have landed roles at **NVIDIA** and other top AI shops primarily because they could speak fluently about fine-tuning. It's the single most impressive thing you can casually drop into an interview answer.

Projects to ship this month

BUILD: Multi-Agent Research Assistant

Use AutoGen or CrewAI to orchestrate agents that find papers, summarize them, and stitch the notes into a literature review. This is the kind of project that takes you from "interesting candidate" to "hire immediately."

BUILD: Document → Excel Parser

An AI Engineer interview I gave actually asked: "Design a system that converts unstructured documents into structured Excel rows." Build this with a parser + LLM pipeline. It's an unfair advantage on your resume.

End-of-month checklist

- Can explain transformer attention, RAG, and fine-tuning in plain English.
- Built and deployed a small RAG app with a real vector DB.
- At least one project that uses tool calling.
- A multi-agent project on GitHub. Posted on LinkedIn with a video demo.

MONTH

4

Ship It

AI Engineering, MLOps & Production

The goal of Month 4

Move from notebooks to production. Learn how to design ML and agentic systems, deploy them, monitor them, and survive real users. By the end of this month you'll have a flagship end-to-end project that recruiters cannot ignore.

MLOps essentials

• **FastAPI** — wrap your model in an HTTP API.
• **Docker** — containerize the service so it runs anywhere.
• **CI/CD** — GitHub Actions for automated testing and deploy.
• **Cloud deploy** — AWS, GCP, or a simple PaaS like Render/Railway.
• **Monitoring** — Prometheus + Grafana, or a hosted equivalent.
• **Versioning** — DVC for data, MLflow or Weights & Biases for experiments.

Open-model serving

Recruiters care about cost and latency. Two tools to know:

• **Ollama** — fastest way to run open models locally for prototyping.
• **vLLM** — production-grade inference server with paged attention. Know the difference between batched and streaming inference.

ML & agentic system design

Even fresher AI Engineer interviews now include a system-design round. Be ready to talk about: data pipelines, feature stores, online vs. batch inference, A/B testing, rollback, retrieval architectures, and how to scale agent loops without burning a thousand dollars a day in tokens.

Awesome System Design Resources (free, comprehensive) — github.com/ashishps1/awesome-system-design-resources
Bookmark this. It's the best free system-design vault on GitHub.

Agentic Design Patterns (free PDF) — github.com/sarwarbeing-ai/Agentic_Design_Patterns
Practice these and you'll ace any agent-design interview.

The flagship project

THIS IS THE PROJECT THAT GETS YOU HIRED

Take an open base model. Fine-tune it on a real-world use case (legal Q&A;, healthcare summaries, finance ratios — pick a domain). Wrap it in a FastAPI service. Containerize with Docker. Serve it with Ollama or vLLM. Add Prometheus metrics. Deploy to a public URL. Write a long-form LinkedIn post explaining the architecture, the tradeoffs you made, and what you'd do differently. **Pin it to your profile.**

End-of-month checklist

• A live, deployed AI service with a public URL.
• Dockerfile, CI/CD, monitoring dashboard, and a clean README.
• You can sketch a system-design diagram from scratch in 15 minutes.
• You're interviewing — and you have offers, or you're close.

PART TWO

The Free Resources Vault

Finding good resources is easy. Finding *free*, high-quality resources is insanely hard. After two years in this field, this is my honest, hand-curated list of what's worth your time.

Foundational courses (Stanford / IIT)

Stanford CS229 — Machine Learning (full) — youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU
Full Coursera ML Specialization substitute. Free.

Stanford CS230 — Deep Learning (Andrew Ng) — youtube.com/playlist?list=PLoROMvodv4rNRRGdS0rBbXOUGA0wjdh1X
Free version of the Coursera Deep Learning Specialization.

Stanford Math for ML — linear algebra, probability, calculus — youtube.com/playlist?list=PLoROMvodv4rOpr_A7B9SriE_iZmkanvUg
Taught by top professors. Be grateful — this is gold.

NPTEL NLP — IIT Madras — nptel.ac.in/courses/106105158
Free, certificate-eligible if you sit the offline exam. Resume gold.

AI Engineer Toolkit — break-into-data — github.com/break-into-data/ai-engineer-toolkit
Curated free courses, end-to-end.

System design (notoriously gate-kept, this one is free)

Awesome System Design Resources — github.com/ashishps1/awesome-system-design-resources
Save this RIGHT NOW. Big-tech designs, FAQ questions, the lot.

People to follow on LinkedIn

Chip Huyen — AI engineering author & legend — linkedin.com/in/chiphuyen
Read every post. Read her books.

Alex Xu — bestselling system design author — linkedin.com/in/alexxubyte
Step-by-step concept breakdowns. Brilliant.

Aditya Rai — daily ML interview questions — linkedin.com/in/aditya-rai-ds
Solve one a day. You'll outclass the average candidate in a month.

Project repositories

AI Project Gallery (Kalyan M45) — by algorithm category — github.com/KalyanM45/AI-Project-Gallery
Classification, CV, RAG, agentic — all free notebooks. Don't copy verbatim — interviewers will spot it.

Re-implement.

Practical Machine Learning — full EDA → modeling notion site — gcolonna.notion.site/practical-machine-learning
The best end-to-end pipeline walkthrough I've found. Free.

Agentic Design Patterns (free PDF book) — github.com/sarwarbeing-ai/Agentic_Design_Patterns

Practice & interview prep

Deep-ML — "LeetCode for ML" — deep-ml.com/problems
Theory and code questions. Math drills. Research-paper-based questions. Companies actually use these.

interview_questions — by role (ML / AI / AI Research) — github.com/aakriti1318/interview_questions
ML, AI, and AI Research engineer interviews are different. This sorts them.

ROLE DIFFERENCES — KNOW THESE BEFORE YOU APPLY

ML Engineer interviews focus on data/feature engineering, model training, experimentation, metrics, performance tuning.
AI Engineer interviews look more like SWE + applied AI: using pretrained LLMs in real systems, not training from scratch.
AI Research Engineer interviews go deep on papers, novel architectures, and math. Pick the role that matches your strengths and prepare accordingly.

PART THREE

The Resume Playbook

When I started, my resume was getting auto-rejected everywhere. No mentor, no clue. Here's the honest, section-by-section playbook that fixed it — and that I now use to review resumes for friends and mentees who go on to land roles in 1–2 weeks.

1. Photo & contact

Add a small professional photo. Resumes without one get rejected at much higher rates than people think. Then include all four: **phone, email, GitHub, LinkedIn**. If you have a competitive coding profile you're proud of (Codeforces, CodeChef, LeetCode), add it.

2. NO summary section

Recruiters hate it. It duplicates information that's already on the page. Skip it and use that real estate for projects.

3. Length & format

One page is ideal. Two pages is the absolute maximum. Mine was 1.5 — that's fine. Use **LaTeX** (Overleaf is free) — it looks professional, it's ATS-friendly when done right, and it's a skill in itself.

4. Education

Include CGPA **only if it's above 8.5**. Below that, omit it — ATS filters auto-cut you. List your degree, college, year, and any relevant coursework.

5. Work Experience (next, before projects)

For ML/AI roles, prior tech internships are essential. If you don't have one yet:

- Learn Python + 5–10 mini Python projects.
- Apply for a remote part-time **Python internship** on LinkedIn / Wellfound. Paid or unpaid, take it.
- Solve ~50% of Striver's DSA Sheet + Blind 75. Apply for SWE internships in parallel.
- Build agentic AI / RAG / fine-tuning projects, then apply for AI Engineer internships directly.
- Aim for at least **2 tech internships** on your resume before going for your first ML role.

6. Projects

This is the section recruiters skim hardest. Two questions you must answer in every bullet: **what did you build, and what was the measurable outcome?** Pull projects from professor-led work, college majors/minors, Kaggle competitions, hackathons, or the Educative.io project gallery. Use HuggingFace for datasets and pre-trained models.

> **THE TRICK THAT LANDED MY FIRST ML INTERVIEW**
>
> I posted a video demo of a project on LinkedIn. A recruiter saw it and DMed me. That became my **TurboML** internship — my first ML role. Post your projects. Always.

7. Skills

Languages first — list them and mark *(basics)* if you're not fluent. Then Python libraries: **NumPy, Pandas, Matplotlib, scikit-learn, PyTorch**. Then **SQL** — non-negotiable. Then **Git**, shell utilities, Linux. Web frameworks (Django/Flask/React) are optional but a plus.

8. Achievements (only if strong)

JEE/SAT/entrance ranks, scholarships, top-percentile awards. If your achievements are weak, skip the section — it's better to say nothing than something underwhelming.

9. Extracurriculars

Short, but include something. Recruiters do read it — they want to know you're a human, not a list of frameworks.

> **REMOTE-JOB TWEAK**
>
> If you're targeting remote roles, add a one-line tag near the top: "Open to remote / async — comfortable with EST/PST overlap." Subtle, but remote hiring managers actively scan for it.

PART FOUR

14 Recruiter-Magnet Projects

Forget Titanic. Forget MNIST. Forget the same sentiment analysis project everyone submits and then wonders why their resume doesn't get shortlisted. Here are 14 projects — mostly ordered easy → hard, but not strictly, just to keep you on your toes — that I personally would shortlist if I saw them on a candidate's resume.

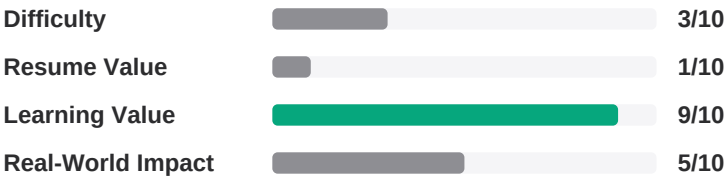
HOW TO READ THE RATINGS

Difficulty — how hard it is to build.
Resume Value — how impressive it looks to recruiters.
Learning Value — how much you'll actually grow.
Real-World Impact — does it solve a real problem?

01 Build Linear Regression From Scratch

Foundations · No scikit-learn, just NumPy and math.

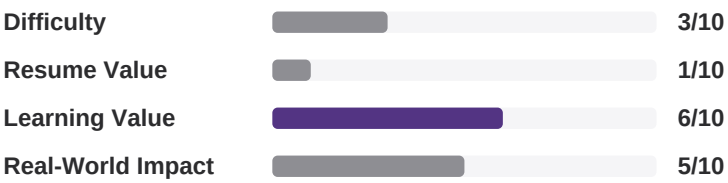
Build it from raw NumPy. The kind of project that saves you at 2 a.m. when a deep model is misbehaving and you need to remember what's actually happening under the hood.



02 Housing Price Predictor

Classic regression · End-to-end ML lifecycle.

Use a Kaggle housing dataset. Practice EDA, feature selection, multiple models, and honest evaluation. Low resume value alone — but high learning value, and a great warm-up.



03 Multi-Agent Job Search System (CrewAI + LangChain)

Agents · Modular task agents that automate your job hunt.

Define modular CrewAI agents — scraper, summarizer, matcher, applier. Wrap with Streamlit. Multi-agent + LLMs + APIs + UI in one project — recruiters love this.



04 Image Classification (Cats vs. Dogs)

Computer Vision · Your first real CNN.

Build a CNN from scratch, then add transfer learning. Document the gap. The reason it's low on resume value is volume — everyone has it. Put a unique twist on it (e.g. a custom dataset) and it becomes a 6/10 instead.



05 AI Travel Planner with A2A Protocol

Agents · Coordinated multi-agent travel assistant.

Use Google's A2A (agent-to-agent) protocol. Real APIs for flights, hotels, attractions. Recruiters specifically look for awareness of A2A in 2026 hiring.



06 Multi-Agent Research Assistant (AutoGen)

Agents · Automate your literature review.

AutoGen + agent collaboration + human-in-the-loop. Reusable across any research-heavy domain. Has genuine startup potential.

Difficulty	**8/10**
Resume Value	**10/10**
Learning Value	**10/10**
Real-World Impact	**10/10**

07 Recommendation System

Classical ML · Collaborative + content-based filtering.

Build collaborative + content-based recommenders. Evaluate with proper offline metrics. Recommenders are everywhere — every e-commerce, content, and social product needs one.

Difficulty	**8/10**
Resume Value	**10/10**
Learning Value	**10/10**
Real-World Impact	**10/10**

08 Language Model From Scratch

Deep Learning · n-grams to a small Transformer.

After this, you actually understand transformer diagrams instead of nodding at them. Resume value is low because it's been done a thousand times — but the learning is irreplaceable.

Difficulty	**10/10**
Resume Value	**1/10**
Learning Value	**10/10**
Real-World Impact	**5/10**

09 Amazon ML Challenge Submission

Hackathon · Real, judged, competitive.

The 2024 challenge required extracting entity values from images. Try RAG first — if it fails, fine-tune. The lesson: try the cheapest method first. Target the 2026 edition. It's a fast track into Amazon.



10 AI Financial Analyst Assistant (DSPy + Flask)

Finance × AI · LLMs over real-time market data.

Pull live stock data, compute ratios and indicators, run reasoning chains, ship in a Flask app. Finance projects scream "HFT-curious" to recruiters at quant and fintech firms.



11 MLOps Pipeline (end-to-end)

Production · Versioning, monitoring, retraining.

The project that lets you legitimately put "MLOps" on your resume. Build the full pipeline — model versioning, monitoring, automated retraining, deployment. This is what companies need but rarely teach in courses.



12 Agentic Dev Workflow (CrewAI + GitHub MCP)

Agents + MCP · Automated GitHub status reports.

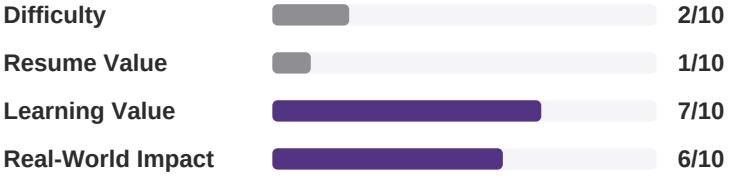
MCP is the new hot signal in AI engineering hiring. Connect CrewAI agents to a GitHub MCP server, fetch issues/PRs/branches, and produce shareable summaries. This single project can land you a top-tier startup interview.



13 YouTube Video Summarizer (Beginner)

LLM Apps · Your literal first AI engineer project.

Pull a transcript, chunk it, summarize. Don't put this alone on your resume — but it's the perfect first repo and a great LinkedIn warm-up post.



14 Document → Excel Parser (LLM Pipeline)

LLM Apps · A real interview question turned project.

I was actually asked this in an AI Engineer interview. Build it. Add OCR fallback for scans, schema validation for outputs, and an evaluator. Pin it to your GitHub.



PART FIVE

Visibility & The Final Push

Build it, then post it.

The single biggest unlock in my career was treating LinkedIn as a public lab notebook. I posted projects, concepts I was learning, demo videos, lessons. None of it was polished at first. It didn't matter. **Visibility compounds.** A recruiter saw one of my project videos and that was how I got my first ML internship at TurboML.

Set up your LinkedIn — yesterday

If you're in college, do this in your first or second year. Don't wait until placements. Once a week, post:

- A project (even a small one). Demo video > screenshot > text-only.
- A concept you just learned, in your own words.
- A snippet of an interview question and how you'd solve it.
- A lesson from something that went wrong.

> **POSTING SCRIPT THAT WORKS**
>
> Hook (1 line) → Context (2 lines) → What I built / learned (3–5 bullets) → What I'd do differently (1 line) → Call to action (1 line). **Always** attach a short demo video or a clean GitHub link.

Cold-email IIT/IIIT/IISc professors

Find professors whose ML/AI lab work overlaps with your interests. Send a 5-line email: who you are, one project you've shipped (link), one paper of theirs you've actually read, and a specific ask ("can I help on X for 2 months unpaid"). Most won't reply. One in twenty will. That one is your career.

Find work on Wellfound

Most people only know LinkedIn. **Wellfound (formerly AngelList)** is, in my opinion, an even better hunting ground for early-career AI Engineers — it's full of US startups actively hiring, with founders who often respond to applications themselves. Several of my mentees got their break this way.

Hackathons & Kaggle

Pick one Kaggle competition every two months. You don't need to win — even a top-25%% finish is a real, defensible resume bullet. The 2026 **Amazon ML Challenge** is specifically worth targeting if you can swing the time — it has a track record of leading directly to Amazon hires.

"You're not behind. You're early. The market is wide open. Now go build, ship, and post."

BONUS

Quick Reference Cheat Sheets

The 4-Month Roadmap, at a Glance

Month	Theme	Output
1	Python · Math · Classical ML	First repo: a small ML project. Andrew Ng cert.
2	Deep Learning · PyTorch · SQL	Image classifier, regression project. Internship applications live.
3	LLMs · RAG · Vector DBs · Agents	Multi-agent project. RAG app. Tool-calling demo.
4	MLOps · Deployment · Production	Live, deployed, monitored AI service. Flagship resume project.

Tools you must know cold

Layer	Tools
Language	Python, SQL, basic shell
Data	NumPy, Pandas, Matplotlib
Classical ML	scikit-learn
Deep Learning	PyTorch (primary), TensorFlow (familiarity)
LLMs / Agents	LangChain, LlamaIndex, CrewAI, AutoGen, DSPy, MCP
Vector DBs	ChromaDB, FAISS, Pinecone (familiarity)
Serving	FastAPI, Ollama, vLLM
Ops	Docker, GitHub Actions, MLflow / Weights & Biases
Cloud	AWS, GCP, or Render/Railway for first deploys

Interview rounds you should expect

Round	What it tests

DSA	LeetCode-style problem solving (medium difficulty bar)
ML / AI fundamentals	Bias-variance, regularization, optimizers, attention, RAG
Coding	Python, NumPy/Pandas, model debugging, sometimes SQL
System design	ML system design or agentic system design (free vs. paid LLMs, latency, cost)
Project deep-dive	Walk through a project on your resume — be ready for 30+ minutes
Behavioral	Why this company, why ML, conflict, learning, ownership

ONE LAST THING

Don't read this and shelve it.

I genuinely want to see you succeed. I've mentored hundreds of people on Topmate for ML Engineer, AI Engineer, and AI Research Engineer roles. I've watched the students who treat this roadmap as a checklist and execute land roles in 4 months. And I've watched the ones who read it twice, nodded, and went back to scrolling still be in the same place a year later.

The market doesn't reward the most talented person. It rewards the one who shipped, shared, and showed up consistently. **Be that person.**

YOUR FIRST 24 HOURS

1. Open a fresh GitHub repo named **ai-engineer-blueprint-{your-name}**.
2. Commit a README with your 4-month plan and weekly goals.
3. Post on LinkedIn: "Starting a 4-month sprint to land an AI Engineer role. Will share progress weekly."
4. Start Programming with Mosh. Tonight.

"If you're happy, I'm happy. See you as an AI Engineer in 4 months."

— Sreemanti

Stay in touch

If you get stuck on anything in this roadmap, reach out. The fastest way to my calendar is a 1:1 session on **Topmate**. You can also DM me on **LinkedIn** or comment on my **YouTube** videos — I read every one.